

ABHINAV KAUSHIK

Senior Data Scientist

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SUMMARY

Senior Data Scientist with 8+ years applying machine learning to real-world data across Healthcare, Telecom and AIOps, with an MSc in Machine Learning and AI. Builds regression and classification models with ensemble methods (XGBoost, LightGBM, Random Forest), alongside time-series forecasting, optimization and statistics, working from SQL, BigQuery and Spark data pipelines to production models served with FastAPI and MLflow. Turns business problems into modelling solutions together with stakeholders. Alongside this, I am building ThinkTree.AI, a non-profit AI learning platform used by NGO teachers to support students with ADHD.

PROFESSIONAL EXPERIENCE

Senior Data Scientist | United Health Group

Bengaluru, India | Nov 2025 - Present

- Deployed a production lab-extraction agent with Claude Opus 4.7 and Amazon Textract, integrated with LIMS, handling about 1,800 lab documents and automating the work behind 800 Jira tickets and PRD tasks.
- Built a production multi-agent LangGraph analytics chatbot for AARP medical plans, answering business questions with PySpark analysis, multi-turn reasoning and live matplotlib and plotly charts, cutting manual analysis by 98%.
- Built automated evaluation and observability pipelines for Question Answering and Summary Agents with LLM evaluators, tracing and failure detection, reducing human-in-the-loop review by 43%.

Senior Machine Learning Engineer | Viavi Solutions

Chennai, India | Sep 2024 - Nov 2025

- Developed an internal RAG-based Project Discovery Assistant in LangGraph combining semantic retrieval with LLM reasoning, reaching a 90% match rate against existing organizational solutions.
- Built an event-correlation engine clustering alarms with Sentence-BERT and DBSCAN to trace the cascading effect of network alarm floods back to the root event, improving semantic alarm classification by 73.5%.
- Prototyped 15+ AI applications for AIOps and GeoPerformance clients from client requirements, with 7 moving into production or paid client work.

Data Scientist | Viavi Solutions

Bengaluru, India | Nov 2021 - Sep 2024

- Deployed a production SLA breach classification model with LightGBM, FastAPI and MLflow, using GPT-3.5 for real-time root-cause analysis and remediation recommendations per breach, reducing downtime by 67%.
- Shipped time-series demand forecasting and anomaly detection on 15+ KPIs across 500+ telecom network devices with Holt-Winters and Isolation Forest, enabling earlier detection and proactive network planning.
- Built a demand-forecasting pipeline on CDR data for the Delta cellular network with LSTM (TensorFlow) and XGBoost regression, improving capacity allocation with a 43% efficiency gain in call quality and SMS delivery.
- Built location-intelligence analytics in Python with GeoPandas, pandas, NumPy, matplotlib and plotly for near-real-time traffic monitoring, used by network teams across LATAM and NAM to spot congestion hotspots.

Data Analyst | Optum

Gurugram, India | Jun 2021 - Nov 2021

- Optimized a scikit-learn Random Forest classifier for member risk stratification with 12+ engineered features, GridSearchCV and 5-fold CV, lifting AUC from 0.74 to 0.81 and recall from 0.62 to 0.73.
- Optimized hospital staff scheduling in Python as a mixed-integer linear program with shift-coverage constraints, and modelled patient wait times with queueing statistics to balance staffing levels against demand.
- Analyzed member plan utilization and renewal behavior across individual and child plan segments using BigQuery, building segmentation queries and reporting pipelines that fed downstream analytics and member risk modelling.

Data Engineer | Nippon Data Systems

Gurugram, India | Nov 2019 - May 2021

- Engineered SQL and PySpark pipelines on large healthcare datasets to automate BAU medical-audit reporting with automated visualizations, cutting manual reporting effort by 95%.
- Wrote complex SQL queries with multi-table joins, CTEs and window functions, delivering ad-hoc reports and analyses for multiple stakeholder teams.
- Built a real-time Kibana dashboard on Elasticsearch for government COVID QRT teams to monitor district-level infection hotspots and coordinate medicine-logistics dispatch, cutting reporting lag by 80%.

Associate Consultant | Saksoft

Gurugram, India | Dec 2017 - Nov 2019

- Built a CatBoost classification model predicting personal-loan applications within 7 days of contact, enabling targeted outreach and customer prioritization.
- Built customer dashboards for financial and product insights with QlikSense, SQL and SAP HANA, increasing user engagement by 40%.

PROJECTS

Think Tree | ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

Courier ETA Prediction Engine

- Built an ETA prediction model for on-demand last-mile logistics, combining order, distance and courier-supply features in LightGBM and XGBoost regression with quantile estimates of late-arrival risk, reducing median prediction error by 17% against a rolling-average baseline.
- Automated retraining and batch prediction with PySpark and Airflow on BigQuery, tracked experiments in MLflow, and evaluated with time-based backtesting.
- **Technologies:** BigQuery, Spark, Python, scikit-learn, LightGBM, XGBoost, MLflow, Airflow

RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

Financial Fraud Detection | Mar 2025

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

Restaurant Feedback Analyzer | Aspect-Based Sentiment Analysis

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

EDUCATION

Master of Science, Machine Learning and AI Liverpool John Moores University	Jan 2021 - Jan 2023
Bachelor of Technology, Information Technology Jamia Hamdard	Aug 2013 - Aug 2017

TECHNICAL SKILLS

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Random Forest, Ensemble Models, Regression, Classification, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Isolation Forest, LSTM, TensorFlow, Keras, PyTorch, SBERT, DBSCAN, Deep Learning

Optimization: Linear and Integer Programming, Scheduling Optimization, Queueing Models

Programming / Data: Python, SQL, PySpark, Pandas, NumPy, Matplotlib, Plotly, GeoPandas, FastAPI, Flask, Hive, Spark SQL, REST API, Pydantic, Unix

Data Platforms: BigQuery, Apache Spark, Databricks, Databricks SQL, Supabase

MLOps / Cloud: MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bamboo, Bitbucket, GCP, Azure, Vercel

AI / GenAI: Generative AI, LLM, RAG, Agentic RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP, Tool Orchestration, Amazon Textract

Evals / Observability: LLM Evals, LLM Evaluators, Agent Tracing, Observability, Elasticsearch, Kibana, Log Analysis

Databases: TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

Search: Apache Solr, Lucene Indexing, Information Retrieval

BI: Power BI, QlikSense, SAP HANA

LANGUAGES

English: Native
German: A1, Pursuing

ADDITIONAL INFORMATION

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies